Learning Module 3: Portfolio Management: An Overview

LOS 3a: describe the portfolio approach to investing

Investors have to ensure their investments **achieve their future needs**. A **portfolio approach** to investment decision-making is important regardless of future financial goals. It enables an investor to create a **diversified** investment portfolio.

Portfolio Diversification

The benefits of a diversified portfolio are best summarized by the anecdotal wisdom of “don’t put all your eggs in one basket.” If a portfolio is too heavily allocated to one individual security, and that security fails for some reason, an investment portfolio can be reduced to zero. Diversification allows investors to spread some of the downside risk associated with any one investment position without necessarily decreasing the expected rate of return.

Reducing Risk

Diversified portfolios have a lower portfolio risk or volatility (as measured by standard deviation) than any individual position within the portfolio. Due to the statistical relationship or interactions between individual portfolio positions (correlation), the overall volatility is lowered by including multiple positions within a portfolio.

Composition Matters

Each security has a historical risk and return profile. By combining securities within a portfolio, we can produce a risk and return profile for the portfolio itself. Through examination of different portfolios – allocating different percentages to the underlying securities – we can determine the portfolio composition that produces the best risk-return profile.

Downside Protection

Portfolio diversification is not an absolute failsafe for investors. A diversified portfolio may not protect an investor from losses during market turmoil. The reduction in risk as a result of portfolio diversification comes about due to the uncorrelated nature of individual portfolio holdings. However, this correlation is not fixed. As was the case during the 2007-2008 financial crisis, previously uncorrelated assets can become correlated when markets are stressed. This is known as contagion, and in 2008, global assets experienced price declines. This proved that portfolio diversification is not as effective as the mathematical models had previously suggested.

The Modern Portfolio Theory

Although adopting a portfolio approach to investing seems intuitive, there is a theory behind it. The diversification concept follows the work of Harry Markowitz's 1952 publication and is known as Modern Portfolio Theory (MPT). The principle concept is that investors should not only hold portfolios but also focus on the relationship among the individual securities within the portfolio. MPT has its limitation but continues to be a cornerstone for portfolio managers.

Question

A portfolio approach to investing provides which of the following benefits?

A. The highest investment returns.

B. Protection against investment losses.

C. A reduction in risk during normal market conditions.

Solution

The correct answer is C.

A diversified portfolio reduces risk without compromising investment returns. However, it does not completely protect an investor against investment losses during times of market turmoil.

LOS 3b: describe the steps in the portfolio management process

Upon determining the type of investment client and their financial goals, a portfolio manager takes a series of steps to ensure the client meets their goals and needs.

1. The Planning Step

Once a portfolio manager has established a client's objectives and constraints, they must develop an investment policy statement (IPS). This written document spells out all the investment objectives and constraints that apply to a client's portfolio. It may also contain a reference to a benchmark. A benchmark can be used to assess investment performance and evaluate whether the objectives have, indeed, been achieved.

2. The Execution Step

The execution step has three stages – asset allocation, security analysis, and portfolio construction.

Asset Allocation

The analyst or portfolio manager determines the economic and capital market expectations for various available asset classes. This analysis may be top-down. In this instance, it starts with a consideration of the macroeconomics or industry environment and an evaluation of the asset classes expected to perform well given the environment. Alternatively, the analysis may be bottom-up. Instead of looking at macroeconomics or industry data, this analysis focuses on company-specific factors. A decision will then be taken on the allocation of assets to the available asset classes. Asset classes can include equities, bonds, cash, real estate, commodities, hedge funds, and private equity.

Security Analysis

An analyst can combine top-down and bottom-up views in selecting individual securities to assess the level of returns and risk. This informs the assignment of a valuation to securities being considered for portfolio inclusion.

Portfolio Construction

A diversified portfolio can be constructed using the investment policy statement (IPS), the desired asset allocation, and security analysis. Besides achieving investment performance, risk management is critical in the portfolio construction process. The IPS will outline a client's risk tolerance, and the portfolio manager must ensure the portfolio is aligned with this risk profile. Once the portfolio manager has chosen securities to buy and the quantities in which to buy them, the transactions will be executed. Often, a specialized trade execution team or external stockbroker executes these transactions.

3. The Feedback Step

After a portfolio manager has constructed a portfolio has been constructed, they need to review and monitor it at an appropriate interval.

Portfolio Monitoring and Rebalancing

Portfolio rebalancing occurs when a portfolio has shifted from the targeted asset allocation due to market movements. If the top-down or bottom-up views change, an individual security or asset class may need to be changed. A change in a client's circumstances may prompt a revision of the IPS and the portfolio.

Portfolio Measurement and Reporting

Portfolio performance must be evaluated to establish whether the client's objectives have been met. Portfolio performance may be assessed in relation to the benchmark set out in the IPS. Following analysis of the performance, it may be determined that the client's objectives have changed. This realization will be factored into the planning and execution steps.

Question

Select the correct sequence of portfolio management steps.

A. Equity valuation, portfolio performance assessment, trade execution.

B. IPS creation, portfolio rebalancing, top-down analysis.

C. IPS creation, portfolio construction, monitoring, and rebalancing.

Solution

The correct answer is **C**.

The portfolio management process must begin with the creation of an investment policy statement in the planning step. This is followed by analysis and portfolio construction in the execution step. Finally, rebalance, performance measurement, and monitoring are carried out in the feedback step.

LOS 3c: describe types of investors and distinctive characteristics and needs of each

The needs of investment clients vary widely, but we can group investors into two broad categories – **individual** and **institutional** investors. Different investors will have varying investment time horizons, tolerance for portfolio risk, income, and liquidity needs.

Individual Investors

Individual investors may be investing either for short-term or long-term goals. A short-term investment goal may be their children’s education or the purchase of a house. Long-term investment goals revolve around providing income for retirement. The implication is that some investors are focused on capital growth and look for those investments with the potential for capital appreciation, while retirees will want income-producing assets. The structuring of a portfolio for an investor will also be dependent on their financial circumstances, such as home ownership, employment prospects, and other financial obligations.

Institutional Investors

There are many different types of institutional investors. Indeed, institutional assets constitute a major portion of the investment market. Pension funds, endowments, charities, banks, insurance companies, investment funds, and Sovereign Wealth Funds (SWF) are all classified as institutional investors. These institutional investors also have different financial objectives.

Endowments and Foundations

The typical objective of an endowment or foundation is to maintain the real (inflation-adjusted) capital value of the fund in perpetuity as well as generate income to provide financial support for its beneficiaries.

Banks

Banks hold deposits and make loans which can lead to excess reserves – in this case, a bank holds more deposits than the loans it has extended. Banks can invest these reserves, which typically have to be held in conservative and liquid assets like fixed-income and money market instruments. The bank's objective is to earn a rate of return in excess of the rate of interest it pays on its deposits.

Insurance Companies

Insurance companies receive premiums from the insurance policies they write. They need to invest these premiums to ensure sufficient funds are available to pay for insurance claims when these arise. As such, their investments are also often conservative in nature and cognizant of the investment time frame over which claims may arise.

Investment Companies

Investment companies manage mutual funds, which are pooled investment vehicles. Mutual funds are seen as an efficient way for individual investors to gain access to a diversified portfolio and benefit from the skills of a professional investment manager. Mutual funds are managed according to the limits and restrictions of their investment mandates.

Sovereign Wealth Funds

SWFs are government-owned investment funds. Some governments operate with the objective of investing the revenues from the country's natural resources (e.g., oil) for the benefit of future generations. On the other hand, others manage state assets.

Question

Excess reserves held by banking institutions are usually invested in:

A. Emerging market equities and other high-growth stocks.

B. Money-market and fixed-income instruments.

C. Real estate and other tangible assets.

Solution

The correct answer is **B**.

Banks invest their excess reserves in conservative and liquid assets such as fixed-income and money-market instruments.

LOS 3d: describe defined contribution and defined benefit pension plans

Employees of both private and public companies often save and invest for retirement via defined contribution (DC) pension plans in which they assume the investment risk. In the case of a defined benefit (DB) pension plan, the responsibility and investment risk fall on the employer. It is up to the employer to provide the defined benefits to employees on retirement.

Defined Contribution Pension Plan

A Defined Contribution (DC) pension plan is an investment vehicle in which the amounts invested, or the contributions that the employee makes to the plan, are defined or specified, but the benefits are not predetermined. The objective of the pension plan is to accumulate wealth by investing a portion of wages while working to provide income during retirement. Unlike a Defined Benefit (DB) pension plan, where the retirement benefits are predetermined based on factors such as salary and years of service, a DC plan places the investment risk on the employee. The employee is responsible for ensuring that their contributions and investment growth are sufficient to provide the desired income upon retirement. The final retirement income in a DC plan depends on the contributions made, investment performance, and the choices made by the employee regarding investment options within the plan.

Defined Benefit Pension Plan

In a DB pension plan, the employer has an obligation to provide certain benefits to employees when they retire. The future benefit is specified or defined. The investment management of DB plans has to consider the timing of its future liabilities or cash flows by assessing the age of its plan members. If, for example, a DB pension plan has a lot of young members, the investment time horizon of the plan may be quite long. The plan investment manager may try to match the cash flow requirements of the plan with cash-flow-producing assets such as bonds. This checks the capacity of portfolio assets to offset portfolio liabilities.

Question

Which of the following *best* describes the investment risk of a DB pension plan?

A. The employer assumes the investment risk.

B. The employee assumes the investment risk.

C. The employer and the employee share the investment risk.

Solution

The correct answer is **A**.

In a DB pension plan, the employer must ensure there are sufficient assets to match the future defined benefits payable to employees upon retirement.

LOS 3e: describe aspects of the asset management industry

Buy-Side vs. Sell-side

Asset managers are typically deemed to be on the buy-side. This ideally means that they buy the products of sell-side firms. Buy-side firms include asset managers, hedge funds, institutional investors, and retail investors. On the other hand, sell-side firms include investment and commercial banks, stockbrokers, and market makers.

Types of Asset Managers

Active vs. Passive Managers

Active managers represent around 80% of the asset management industry. Through fundamental and quantitative research, they attempt to outperform benchmarks, such as the S&P 500. On the contrary, passive management simply tries to replicate the returns of a market index.

Traditional vs. Alternative Asset Managers

Traditional management usually focuses on long stocks and bonds to create diversified client portfolios. In contrast, alternative management uses leverage, derivatives, long-short strategies, etc., to either outperform a predetermined index or to create a return that is uncorrelated to the market.

Lately, there has been a blurred line between traditional and alternative management. Many traditional managers have introduced higher-margin alternative products to clients.

Ownership Structure

Most asset managers are privately-owned firms. However, some publicly-traded asset management firms also exist. These often offer services such as insurance and/or banking services.

Asset Management Industry Trends

Growth of Passive Management

Passive management is gaining currency in the asset management industry. The two key reasons for this are the low fees charged for passively-managed funds and the fact that active managers are having a hard time beating indices in increasingly-efficient markets.

Big Data

Nowadays, algorithms are much faster at analyzing earnings and economic news than humans. This opens the door to short-term trading. In fact, many asset managers are now using machine-learning techniques to help process data.

In order to generate alpha, asset managers are trying to discover data with predictive potential faster than fellow market participants.

Robo-Advisors

In 2017, robo-advisers managed an estimated USD 180 billion in assets. The main advantages of robo-advisors are that:

- They seem to attract a **younger crowd** of investors.
- They charge **lower fees** because of the scalability of the technology.
- They provide a **low barrier to entry** to other firms, such as tech firms, to enter the lucrative asset-management industry.

LOS 3f: describe mutual funds and compare them with other pooled investment products

Separately Managed Accounts

SMAs, also referred to as managed accounts, wrap accounts, or individually managed accounts, are portfolios managed exclusively for the investor according to their investment, tax preferences, and requirements. The investor owns the underlying assets directly, unlike a mutual fund. Due to the individually tailored nature of SMAs, the minimum investment amount is significantly higher than that of a mutual fund, and institutional investors use them.

Many investors choose to participate in a pooled investment vehicle rather than assemble a portfolio of securities by themselves. There are several types of pooled investment vehicles. Mutual and exchange-traded funds (ETFs) tend to have low minimums, while hedge funds and private equity funds may require large investment amounts.

Pooled Investments

Mutual Funds

The value of a mutual fund is referred to as the Net Asset Value (NAV). It is computed based on the closing price of the fund's underlying securities. Each investor owns a number of shares in the fund, which represents a pro-rata claim on the value of the mutual fund.

Open-end Fund

An open-end mutual fund will accept new investor inflows and issue new investors shares in the mutual fund priced at the NAV of the fund at the time of investment. Investors can also sell their mutual fund shares at the prevailing NAV. Therefore, the total number of shares at the disposal of the mutual fund will change depending on its net inflows or outflows.

The portfolio manager of an open-end fund has to manage the cash inflows and outflows. They may, in fact, have to liquidate fund assets to meet redemption requests. Otherwise, they may feel pressure stemming from the demand for more investment opportunities when there are large inflows to the fund. The structure makes it easy for the mutual fund to grow in size by attracting investor assets.

Closed-end Fund

A closed-end fund will not create new shares when a new investor wants to buy shares. Instead, an existing investor will have to sell their shares to the new investor. The total number of shares in issue is fixed. Transactions do not necessarily occur at the NAV of the fund but may be at a premium or discount to NAV.

The portfolio manager of a closed-end fund does not have to manage the cash inflows and outflows. Closed-end funds tend to attract fewer investor assets and only make up a small portion of the mutual fund universe.

Load and No-load Funds

Mutual funds can also be classified as load or no-load funds. A load fund charges investors a sales charge fee to buy, hold or sell shares in the fund. Retail brokers usually sell these funds. The brokers may receive a portion of the fee as a commission. These types of funds are increasingly becoming less popular.

A no-load fund does not charge a transaction-based fee but an annual fee based on a percentage of the fund's NAV.

Types of Mutual Funds

Mutual funds are broadly classified according to the type of underlying assets they invest in.

Money Market Funds

Money market funds are often seen as substitutes for bank deposits. However, they are not insured in the same way. Therefore, there is some degree of risk over a bank deposit. Money market funds are either taxable or tax-free. Taxable money market funds invest in short-term corporate and federal government debt. Tax-free funds invest in short-term state and local government debt.

Bond Mutual Funds

Bond mutual funds invest in individual bonds and, occasionally, preference shares. A key difference between bond funds and money market funds is the maturity of the underlying bonds. Money market funds may hold positions with an overnight maturity. They rarely last longer than 90 days. A bond fund holds positions with maturities between 1 and 30 years. They also hold bonds of various credit ratings.

Stock Mutual Funds

Stock or equity mutual funds have the most assets under management globally. They can either be actively or passively managed. A passive fund is designed to track a particular index through a buy-and-hold strategy. On the other hand, an actively managed fund comprises equity securities selected by the portfolio manager seeking outperformance. The fees on actively managed funds are higher than those on passive funds and tend to be traded more actively. This more active trading has a tax implication. It attracts higher taxes relative to an index fund.

Hybrid or Balanced Funds

Hybrid or balanced funds invest in both equities and bonds. These funds are gaining popularity as lifecycle funds that target a particular retirement date become more sought after. A lifecycle fund tilts the mix of equities and bonds as the time for retirement draws near.

Other Pooled Investments

In addition to mutual funds, other pooled investments, such as Exchange-Traded Funds (ETFs), separately managed accounts (SMAs), hedge funds, buyout funds, and venture capital funds, are available to investors.

Exchange-traded Funds

ETFs track a basket of securities decided upon by the sponsor. The sponsor interacts with institutional investors who deposit the securities basket with the sponsor and receive creation units in the ETF in return. These units can then be sold to the public by the institutional investor. The institutional investor can also return their units to the plan sponsor in exchange for the securities basket. This creation and redemption mechanism helps ensure the ETF units are priced close to the NAV.

When an investor buys an index fund, the investor buys the shares directly from the fund. However, when an investor buys an ETF, the investor buys the units (shares) from other investors in the same way one buys trading equity securities. ETFs are priced throughout the trading day, and the purchase of an ETF share in the open market may not be conducted at the NAV but at a price that represents investor demand at the time. Under normal market conditions, this is usually close to the NAV.

ETF expenses tend to be lower than index funds, but brokerage is incurred when transacting. Dividends that arise in an ETF are paid out directly to the shareholders. Index funds and mutual funds tend to reinvest dividends.

The minimum investment amount for ETFs tends to be smaller than that for mutual funds, and investors may choose to buy a single share of the ETF.

Hedge Funds

Hedge fund strategies tend to be more complex than those of mutual funds. They can make use of leverage and extensive derivative positions. Many hedge funds are more loosely regulated than mutual funds. However, to be exempted from regulations, they may not market themselves to the general public.

The minimum investment amounts are usually high – millions of dollars – and hedge funds may impose liquidity restrictions. This means that investors have to commit to retaining and maintaining their investments for a particular period of time.

Buyout and Venture Capital Funds

Both buyout and venture capital funds invest in equity positions. Buyout funds aim to buy all the shares of a public company, thereby occasioning the privatization of the company. Often, large amounts of debt are issued in order to buy all the shares. This is known as a leveraged buyout (LBO). The intention is to use the company's cash flow to pay down the debt and restructure the company. This makes the restructured operation suitable for an initial public offering (IPO) or sale to another company, thus providing an exit to investors.

Venture capital (VC) funds do not buy established companies but finance companies in a start-up phase. In addition to financing, VC funds offer close oversight and management input. As with buyout funds, the intention is to list or sell the funded company in a finite and relatively short time to create an exit for investors.

Question

Which option correctly represents the characteristics of ETFs and index funds?

A. Index funds are priced throughout the trading day like equity securities.

B. When investors purchase an ETF, they pay a brokerage fee.

C. ETFs are always priced at the net asset value (NAV) at which investors transact at the closing price a day.

Solution

The correct answer is **B**.

ETFs trade like equity securities throughout the trading day. When buying ETF shares, investors pay a brokerage amount.

Option C is incorrect. ETFs are usually priced close to NAV but may be priced at a premium or discount, given market forces (supply and demand).

Option A is incorrect. Index funds are priced at NAV, do not attract brokerage fees, and are transacted at the closing price on the day.